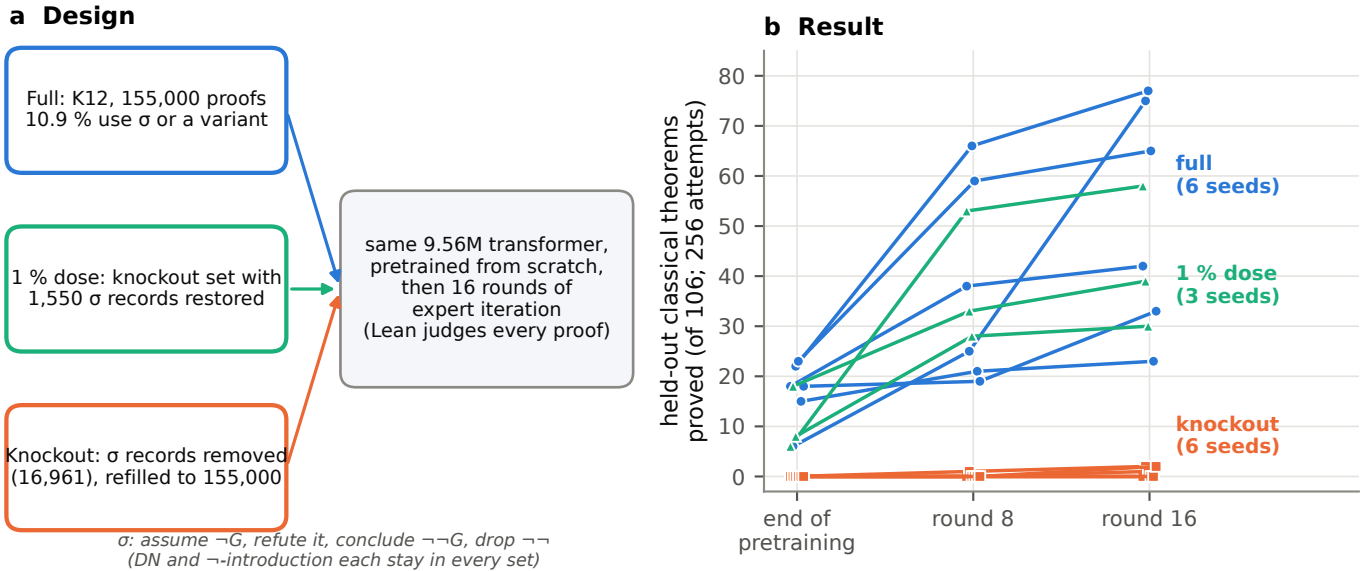
a Design
Full: K12, 155,000 proofs
10.9 % use σ or a variant
1 % dose: knockout set with
1,550 σ records restored
Knockout: σ records removed
(16,961), refilled to 155,000
same 9.56M transformer,
pretrained from scratch,
then 16 rounds of
expert iteration
(Lean judges every proof)
σ: assume ¬G, refute it, conclude ¬¬G, drop ¬¬
(DN and ¬-introduction each stay in every set)
b Result
held-out classical theorems
proved (of 106; 256 attempts)
0
10
20
30
40
50
60
70
80
end of
pretraining
round 8
round 16
full
(6 seeds)
1 % dose
(3 seeds)
knockout
(6 seeds)